

Infosys – Data Engineer Interview Questions & Answers

1. Explain Data Lake vs Data Warehouse with examples.

- **Data Lake:** Stores raw, semi-structured, and unstructured data at scale. Schema-on-read. Example: Azure Data Lake storing IoT sensor data.
 - **Data Warehouse:** Stores structured, cleaned, and processed data. Schema-on-write. Example: Azure Synapse storing sales fact and dimension tables.
👉 *Real-world:* Raw clickstream logs in Data Lake → transformed into structured fact/dim tables in Synapse for BI dashboards.
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2. What is a Star Schema? When do you use it?

- **Star Schema:** Central fact table linked to multiple dimension tables. Easy to query, widely used in OLAP systems.
 - Used for reporting and BI dashboards due to its simplicity and fast joins.
👉 *Example:* Sales fact (amount, quantity) + dimensions (Customer, Product, Time).
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3. How do you handle Slowly Changing Dimensions (SCD) in ETL pipelines?

- **Type 1:** Overwrite old value (no history).
 - **Type 2:** Insert new row with versioning (maintains history).
 - **Type 3:** Add new column for limited history.
👉 *Real-world:* Customer address changes → SCD Type 2 keeps old + new record with effective dates.
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4. What are Delta Tables in Databricks?

- Storage layer built on Parquet with **ACID transactions**.
 - Features: Schema enforcement, Time Travel, Z-ordering, Merge/Upserts.
👉 *Use case:* Updating daily sales without reloading full dataset.
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5. Explain different types of partitioning in Spark.

- **Hash Partitioning:** Based on hash of a key.
- **Range Partitioning:** By value ranges (e.g., dates).

- **Round Robin:** Even distribution.
 - **Custom Partitioning:** User-defined.
👉 *Example:* Partition sales data by order_date to speed queries.
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6. How does PySpark handle data shuffling?

- Shuffling = redistributing data across partitions (during joins, groupBy, distinct).
 - Costly operation → involves disk/network I/O.
👉 *Optimization:* Use broadcast joins, partition pruning, avoid unnecessary shuffles.
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7. What is the difference between persist() and cache() in Spark?

- **cache():** Stores data in memory only (default = MEMORY_ONLY).
 - **persist():** More control → memory, disk, serialized options.
👉 Use cache for repeated use of small data; persist for larger datasets.
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8. How do you handle schema evolution in a data pipeline?

- Use formats supporting schema evolution (Delta, Parquet, Avro).
 - Add new columns with defaults.
 - Backward compatibility checks in ADF/Databricks.
👉 *Real-world:* Adding new column email to customer table without breaking pipeline.
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9. Explain orchestration in Azure Data Factory (ADF).

- Orchestration = scheduling and managing execution flow of activities.
 - Uses **pipelines, triggers, dependencies, parameters**.
👉 *Example:* Ingest → Transform → Load → Notify, all automated in ADF.
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10. What is a pipeline parameter in ADF, and how is it used?

- Dynamic input values passed to pipelines.

- Used for reusability (e.g., file path, table name).
👉 *Example:* Parameterize source path for daily ingestion of multiple files.
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11. Describe common data ingestion patterns.

- **Batch ingestion:** Scheduled, periodic loads (e.g., daily CSV).
 - **Streaming ingestion:** Real-time (e.g., Kafka/Event Hub).
 - **CDC-based ingestion:** Only changed records.
👉 *Example:* IoT sensor → Event Hub → Spark Structured Streaming.
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12. What is the difference between Lakehouse and Data Warehouse?

- **Lakehouse:** Combines raw + structured + ML in one (e.g., Databricks).
 - **Data Warehouse:** Strictly structured for BI (e.g., Synapse).
👉 *Example:* Lakehouse used for ML feature engineering + BI; DWH used for curated reporting.
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13. Explain Azure Synapse dedicated vs serverless SQL pool.

- **Dedicated pool:** Provisioned, reserved compute, predictable cost.
 - **Serverless pool:** On-demand, pay-per-query.
👉 Use dedicated for large workloads, serverless for ad-hoc queries.
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14. How do you optimize PySpark jobs?

- Use **partition pruning, broadcast joins, persist/cache, coalesce/repartition, optimize shuffle**.
 - Use **Delta** instead of raw files.
👉 Profile with Spark UI to find bottlenecks.
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15. What are triggers in ADF?

- Mechanisms to start pipelines.
- Types: **Schedule, Tumbling Window, Event-based, Manual**.
👉 *Example:* Trigger pipeline when new file lands in ADLS.

16. Explain the difference between Linked Service and Dataset in ADF.

- **Linked Service:** Connection to data source (e.g., SQL DB, ADLS).
- **Dataset:** Data structure representing a table/file in that source.
👉 Linked Service = connection, Dataset = schema.

17. What is a surrogate key in data modeling?

- Artificial primary key (e.g., auto-increment ID).
- Avoids dependency on natural keys (which can change).
👉 *Example:* customer_id surrogate key in dimension table.

18. How do you handle large data ingestion in ADF?

- Use **parallelism, partitioned copy, PolyBase, staging in Blob/ADLS**.
- Optimize Integration Runtime.
👉 *Example:* Load 10M+ rows from SQL to Synapse using PolyBase.

19. Explain CDC (Change Data Capture) implementation in Azure.

- **SQL CDC:** Tracks inserts, updates, deletes.
- **ADF Mapping Data Flows / Databricks Merge:** Apply only changed data.
👉 *Example:* Incremental load of customer table using last_modified_date.

20. What is schema-on-read vs schema-on-write?

- **Schema-on-read:** Raw data, schema applied at query time (Data Lake).
- **Schema-on-write:** Schema enforced before storing (DWH).
👉 *Example:* JSON logs in ADLS vs SQL tables in Synapse.

21. Explain optimization techniques in Data Lake queries.

- Partitioning, Bucketing, Z-ordering.
- File compaction (avoid small files).

- Use Delta for indexing.
👉 *Example:* Optimize sales queries by partitioning on order_date.
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22. How do you implement error handling in ETL workflows?

- Use **try-catch** in Databricks.
 - Use **On Failure** activities in ADF.
 - Log errors to monitoring table / Log Analytics.
👉 *Example:* Retry failed copy activity automatically.
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23. What is Z-ordering in Delta Lake?

- Multi-dimensional clustering to speed up queries.
 - Stores related rows closer together.
👉 *Example:* Querying sales by region and date improves performance.
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24. Explain the concept of Medallion Architecture.

- **Bronze:** Raw data.
 - **Silver:** Cleaned, conformed.
 - **Gold:** Curated, aggregated.
👉 *Example:* IoT → Bronze (raw JSON) → Silver (structured parquet) → Gold (aggregates for BI).
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25. What is a cluster in Databricks?

- A set of VMs to run Spark jobs.
 - Types: Job clusters, all-purpose clusters.
👉 Provides distributed compute environment.
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26. Explain job cluster vs all-purpose cluster.

- **Job Cluster:** Created per job, auto-terminated. Cost-effective.
- **All-purpose Cluster:** Shared, interactive, for notebooks & dev.
👉 Use job clusters for production ETL; all-purpose for dev/testing.

27. What is the difference between DataFrames and RDDs?

- **RDDs:** Low-level, no schema, more control, less optimized.
 - **DataFrames:** Schema-aware, SQL-like, Catalyst Optimizer support.
👉 Prefer DataFrames for most ETL/SQL; RDDs for low-level ops.
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28. How do you secure a Data Lake?

- Use **RBAC/ACLs, Azure AD authentication, Encryption (at rest + transit)**.
 - Use **firewall + private endpoints**.
👉 *Example:* Secure ADLS Gen2 with Entra ID + managed identities.
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29. Explain Spark job execution workflow.

- **Driver program** → DAG → Stages → Tasks → Executors.
 - Lazy evaluation: transformations build DAG, action triggers execution.
👉 Monitored in Spark UI.
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30. How do you monitor ADF pipelines effectively?

- Use **ADF Monitor Hub**.
 - Enable **integration with Log Analytics & Azure Monitor**.
 - Configure **alerts & retries**.
👉 *Example:* Email alert on pipeline failure at 2 AM.
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NTT Data – Data Engineer Interview Questions & Answers

1. Explain ETL vs ELT.

- **ETL (Extract → Transform → Load):** Data is extracted from sources, transformed in a staging area, then loaded into target (e.g., Data Warehouse).
- **ELT (Extract → Load → Transform):** Data is loaded raw into the target system (like Synapse or BigQuery), then transformed inside it.
👉 *Example:* ETL with Informatica to SQL DW; ELT with ADF copying raw data to Synapse, then transforming using SQL.

2. What are best practices for building robust ETL pipelines?

- Parameterize pipelines for reusability.
- Use incremental loads (CDC) instead of full refresh.
- Implement error handling & retries.
- Apply logging & monitoring.
- Ensure data validation and quality checks.
👉 *Real-world:* ADF pipeline with parameterized source paths and logging in Log Analytics.

3. How does Spark manage memory internally?

- Divides memory into **execution memory** (for shuffles, joins, aggregation) and **storage memory** (for caching, persisting).
- Uses **Unified Memory Manager**.
- Evicts cached data if execution needs more memory.
👉 *Optimization:* Tune spark.executor.memory and caching strategy.

4. Explain broadcast join in PySpark.

- Small dataset is broadcasted to all executors to avoid shuffles.
- Best when one dataset is small (few MBs).
👉 *Example:* Joining sales fact (huge) with country lookup (small).

5. What are Azure Key Vault secrets, and how are they integrated in ADF?

- **Azure Key Vault:** Securely stores secrets, passwords, keys.
- In ADF, you can link a Key Vault and reference secrets dynamically in Linked Services.
👉 *Example:* Store SQL password in Key Vault, call it from ADF Linked Service.

6. Explain concepts of PolyBase in Synapse Analytics.

- PolyBase allows querying external data (Blob/ADLS, Hadoop) using T-SQL without importing.
 - Parallel load into Synapse tables.
👉 *Example:* Load 1 TB parquet files from ADLS into Synapse fact table using PolyBase.
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7. What is Delta Lake Time Travel?

- Feature to access earlier versions of Delta tables using version number or timestamp.
 - Helps audit, rollback, reproduce experiments.
👉 *Example:* Querying yesterday's version of sales table to debug issues.
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8. How do you implement partition pruning in Spark?

- Spark reads only necessary partitions instead of full dataset.
 - Achieved by filtering on partition keys.
👉 *Example:* Partitioned by year, query with WHERE year=2023 → only that partition scanned.
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9. Explain Azure Event Hub and its use in streaming.

- **Event Hub:** Big data streaming platform, event ingestion service.
 - Supports millions of events/sec, integrates with Spark, Functions, ADF.
👉 *Example:* IoT sensors streaming to Event Hub → Spark Structured Streaming for processing.
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10. How do you design a fault-tolerant data pipeline?

- Use retries and dead-letter queues.
 - Store checkpoints for restarts.
 - Idempotent writes (avoid duplicates).
 - Monitor failures with alerts.
👉 *Example:* Streaming pipeline with Spark + checkpointing + DLQ in Event Hub.
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11. Explain different types of joins in Spark SQL.

- Inner join
- Left/Right Outer join
- Full Outer join
- Cross join
- Semi join
- Anti join

👉 *Example:* Customer table LEFT JOIN Orders to get all customers, including those without orders.

12. How does Spark handle skewed data?

- Skew = uneven distribution of keys, causes slow tasks.
- Fixes: Salting keys, Broadcast joins, Skew join hints.

👉 *Example:* One customer has 90% orders → add random salt to distribute load.

13. How do you optimize ADF Copy Activity?

- Enable parallel copy.
- Use PolyBase for large loads.
- Use appropriate Integration Runtime (Self-hosted vs Azure IR).
- Tune batch size.

👉 *Example:* Copying billions of rows from SQL Server to ADLS using PolyBase.

14. What is the difference between Lake Database and SQL Database?

- **Lake Database (in Synapse):** Schema-on-read, tables mapped on top of files in Data Lake.
 - **SQL Database:** Structured, schema-on-write.
👉 *Use case:* Lake Database for big semi-structured data, SQL DB for transactional workloads.
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15. Explain the concept of data lineage.

- Tracks data origin, transformations, and flow across systems.
 - Helps in debugging, auditing, compliance (GDPR).
👉 *Tool: Azure Purview / Microsoft Fabric Data Catalog.*
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16. What is the difference between batch and streaming ETL?

- **Batch:** Periodic loads (daily/weekly). Latency = minutes/hours.
 - **Streaming:** Continuous, near real-time. Latency = seconds.
👉 *Example: Batch = daily sales load; Streaming = fraud detection in transactions.*
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17. Explain Slowly Changing Dimension Type 2.

- Maintains history by inserting a new row with surrogate key and effective dates.
👉 *Example: Customer moved from NY → LA → new record inserted with validity dates.*
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18. What are primary performance bottlenecks in Spark jobs?

- Data skew
 - Excessive shuffling
 - Small file problem
 - Insufficient partitions or executors
 - Inefficient joins
👉 *Solution: Optimize partitions, cache, broadcast joins, file compaction.*
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19. How do you secure an ADF pipeline?

- Use Managed Identity for authentication.
 - Store secrets in Key Vault.
 - Apply RBAC in ADF.
 - Enable network security (private endpoints).
👉 *Example: Only authorized app can trigger pipeline.*
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20. What is the role of Databricks Unity Catalog?

- Central governance for data & AI assets.
 - Manages access control, auditing, lineage across workspaces.
👉 *Use case:* Define table-level permissions across multiple Databricks workspaces.
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21. How do you manage schema drift in ADF?

- Enable “Auto Mapping” in Copy/Data Flow.
 - Use Data Flows with schema drift enabled.
 - Store unmapped columns in JSON/Blob for audit.
👉 *Example:* Extra column added in source → pipeline adapts without failure.
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22. Explain deduplication techniques in Spark.

- dropDuplicates() function.
 - Window functions (ROW_NUMBER, RANK).
 - Distinct.
👉 *Example:* Keep latest record per customer_id using row_number over timestamp.
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23. What is Azure Purview?

- Data governance & catalog tool.
 - Provides **data discovery, classification, lineage, glossary**.
👉 *Example:* Scan Synapse & ADLS to track sensitive PII data.
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24. Explain how checkpointing works in Spark streaming.

- Saves metadata about processed data.
 - Allows recovery after failure without reprocessing.
👉 *Example:* Kafka → Spark Streaming with checkpointing in ADLS.
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25. What is a data ingestion strategy for IoT devices?

- Use streaming services (Event Hub, IoT Hub, Kafka).
 - Store raw data in Data Lake (Bronze).
 - Process via Spark/Databricks → curated layers.
- 👉 *Example:* Smart meters sending readings every 5 seconds.
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26. How do you handle incremental loads in pipelines?

- Capture changes with CDC or last modified date.
 - Use MERGE in Delta/Synapse.
- 👉 *Example:* Load only yesterday's modified orders, not full table.
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27. Explain the difference between external and managed tables in Databricks.

- **Managed table:** Databricks manages storage + metadata. Drop = deletes data + metadata.
 - **External table:** Only metadata managed, data remains in external storage.
- 👉 Use external for shared data, managed for internal.
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28. How do you design for scalability in Data Lake?

- Partitioning & compaction.
 - Delta format with Z-ordering.
 - Use auto-scaling clusters.
 - Avoid small files.
- 👉 *Example:* Multi-TB log data optimized with partitioning by date.
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29. Explain different methods of data encryption in Azure.

- Encryption at rest: Storage Service Encryption (SSE).
 - Customer-managed keys (CMK) in Key Vault.
 - Encryption in transit: TLS/HTTPS.
- 👉 *Example:* Encrypt ADLS Gen2 with CMK stored in Key Vault.
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30. How do you use Data Factory integration runtime types?

- **Azure IR:** Cloud-based, for data within Azure.
- **Self-hosted IR:** For on-premises to cloud movement.
- **Managed VNET IR:** For secure data movement with private endpoints.
👉 *Example:* Use SHIR for moving Oracle on-prem → ADLS.

HashedIn by Deloitte – Data Engineer Interview Questions & Answers

1. Explain Snowflake schema vs Star schema.

- **Star Schema:** Central fact table linked directly to dimensions. Simple, fast for queries.
 - **Snowflake Schema:** Dimensions are normalized into sub-dimensions. Reduces redundancy, but queries are more complex.
👉 *Example:* In retail, Star schema = Sales fact + Customer/Time/Product dimensions. Snowflake = Product dimension further split into Product → Category → Department.
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2. How do you design a cloud-native data platform?

- Use scalable cloud services: Storage (ADLS/S3/GCS), Compute (Databricks/EMR/BigQuery), Orchestration (ADF/Airflow), Security (Key Vault/IAM).
 - Ensure modularity, scalability, cost optimization, and security.
👉 *Example:* IoT ingestion → Event Hub → Databricks processing → Delta Lake → Synapse for BI.
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3. What are Delta Live Tables?

- A managed ETL framework in Databricks for building **reliable, declarative pipelines**.
 - Features: Auto-handles dependency ordering, schema enforcement, incremental updates.
👉 *Use case:* Define gold/silver tables declaratively, and Databricks manages refresh automatically.
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4. Explain the difference between Streaming and Batch pipelines.

- **Batch:** Processes data in chunks (daily/weekly). Higher latency.
 - **Streaming:** Processes continuous flow, near real-time. Low latency.
👉 *Example:* Batch = monthly financial reporting. Streaming = fraud detection in transactions.
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5. How do you handle late-arriving data in Spark?

- Use **watermarking** in Structured Streaming.
 - Keep a tolerance window (e.g., 10 mins).
 - Use Delta Lake Merge for updates.
👉 *Example:* Streaming events arriving 5 mins late → still processed within watermark.
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6. Explain PolyBase and its advantages.

- PolyBase enables querying external data sources (Blob/ADLS/Hadoop) with T-SQL in Synapse.
 - Advantages: High performance parallel loads, avoids staging, works directly with big data files.
👉 *Use case:* Load raw parquet files into Synapse without moving them.
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7. How do you implement CDC in Databricks?

- Use **Delta Lake MERGE** to apply changes.
 - Capture insert, update, delete from source (via ADF or Kafka).
👉 *Example:* Merge daily sales updates into Delta table using WHEN MATCHED/NOT MATCHED.
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8. Explain how ADF and Synapse integrate.

- ADF handles **orchestration** (ingestion, movement, transformation).
 - Synapse provides **analytics and storage** (dedicated pools, serverless queries).
👉 *Example:* ADF ingests raw files → Synapse external tables for reporting.
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9. What is a cluster policy in Databricks?

- A governance feature to enforce cluster configurations (e.g., max node size, auto-termination).
- Ensures compliance, cost control, and consistency.

👉 *Example:* Limit dev clusters to 2 nodes.

10. How do you handle slowly changing dimensions in a Lakehouse model?

- Use Delta Lake **MERGE** for SCD Type 1 & 2.
- Store current and historical values in Silver/Gold tables.

👉 *Example:* Customer address change → insert new version with validity dates.

11. Explain auto-scaling in Databricks clusters.

- Clusters can automatically scale **up/down workers** based on workload.
- Prevents over-provisioning and reduces cost.

👉 *Example:* Scale from 2 nodes to 10 during ETL peak load.

12. What is Azure Managed Identity?

- Azure AD-based identity assigned to resources.
- Eliminates need for passwords/secrets.

👉 *Example:* ADF pipeline authenticating to ADLS using Managed Identity.

13. How do you optimize joins in Spark?

- Use broadcast joins for small tables.
- Repartition on join keys.
- Avoid skew with salting.
- Cache reused data.

👉 *Example:* Broadcast small lookup dimension during fact joins.

14. Explain Spark DAG execution.

- DAG = Directed Acyclic Graph of transformations.
 - Spark builds DAG → splits into **stages** → tasks → executed on executors.
👉 Lazy evaluation: DAG built first, execution triggered by an action.
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15. What is a surrogate key vs natural key?

- **Surrogate Key:** Artificial (auto-increment ID).
 - **Natural Key:** Business-defined (e.g., SSN, email).
👉 Surrogate key preferred in data warehouses as natural keys can change.
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16. How do you monitor pipeline failures?

- ADF: Monitor Hub, Log Analytics, alerts.
 - Databricks: Job UI, cluster logs, logging to storage.
👉 *Example:* Configure email alert on failure with Log Analytics.
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17. Explain Z-Ordering and Bloom filters.

- **Z-Ordering:** Multi-dimensional clustering for faster queries.
 - **Bloom Filters:** Indexing technique for faster search in large datasets.
👉 *Example:* Speeding queries on customer_id using Bloom filters.
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18. What is data skew, and how do you fix it?

- Skew = uneven data distribution across partitions.
 - Fix with salting keys, repartitioning, broadcast joins.
👉 *Example:* Popular product causing 80% of sales rows in one partition → fix with key salting.
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19. Explain Medallion Architecture.

- Bronze = Raw data.
- Silver = Cleaned/structured data.
- Gold = Aggregated/curated data.
👉 Industry standard in Databricks for scalable pipelines.

20. How do you integrate Kafka with Spark Streaming?

- Use Spark Structured Streaming Kafka connector.
- Consume Kafka topics in micro-batches.
- Apply transformations → write to Delta/Synapse.
👉 *Example:* Stream website logs → Spark Streaming → Delta Lake.

21. What is schema evolution in Delta Lake?

- Delta allows adding/removing columns with **ALTER TABLE** or auto-merge schema.
- Ensures backward compatibility.
👉 *Example:* Add phone_number to customer table without breaking queries.

22. How do you orchestrate multiple pipelines in ADF?

- Use dependency chaining (Wait, If, Until).
- Execute pipelines within pipelines using **Execute Pipeline activity**.
👉 *Example:* Data ingestion → Transformation → Reporting pipelines linked sequentially.

23. Explain data governance in Azure.

- Implement RBAC, encryption, monitoring.
- Use Azure Purview for catalog & lineage.
- Use Key Vault for secrets.
👉 *Goal:* Ensure security, compliance (GDPR/CCPA).

24. What are different stages of Spark job execution?

- DAG creation → Stages → Tasks → Executors.
 - Stages divided by shuffle boundaries.
👉 Monitored in Spark UI.
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25. How do you automate cluster creation in Databricks?

- Use **Databricks REST API** or **Terraform**.
 - Define cluster configs (node size, scaling).
👉 *Example:* CI/CD pipeline creates clusters automatically for jobs.
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26. Explain data versioning strategies.

- Use Delta Lake Time Travel.
 - Maintain snapshots in Data Lake.
 - Tag versions for ML experiments.
👉 *Example:* Keep multiple versions of sales data for auditing.
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27. How do you optimize storage costs in Data Lake?

- Use Parquet/Delta (compressed).
 - Archive cold data to cheaper storage tier.
 - Compact small files.
👉 *Example:* Move 1-year-old data to archive tier in ADLS.
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28. Explain Azure Synapse vs Databricks Lakehouse.

- **Synapse:** SQL-based DWH for analytics, BI.
 - **Databricks Lakehouse:** Combines Data Lake + ML + BI.
👉 Use Synapse for reporting; Databricks for ML + advanced ETL.
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29. How do you validate pipeline output data?

- Apply row counts, hash checks, schema validation.
 - Create test cases for transformations.
👉 *Example:* Validate sales totals in Silver layer = sum of Bronze layer records.
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30. Explain the importance of partition strategy.

- Correct partitioning reduces query cost and improves performance.

- Avoid too many small partitions.
- Partition by high-cardinality fields (date, region).
👉 *Example:* Partitioning orders by order_date speeds up time-based queries.

Capgemini – Data Engineer Interview Questions & Answers

1. Explain your experience with building ETL pipelines.

- Built pipelines using **ADF + Databricks** for ingesting raw data (CSV/JSON/SQL).
 - Applied transformations (joins, aggregations, deduplication).
 - Loaded curated data into Synapse/Data Lake for BI.
👉 *Example:* Daily sales ETL → raw → cleaned → aggregated → Power BI reports.
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2. What are the main differences between Data Lake and Data Warehouse?

- **Data Lake:** Stores raw, semi/unstructured data, schema-on-read. Example: ADLS.
 - **Data Warehouse:** Stores structured, curated data, schema-on-write. Example: Synapse.
👉 Lake = flexible, cheaper; DWH = optimized for analytics.
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3. How do you implement Slowly Changing Dimensions (SCD) in ETL?

- **Type 1:** Overwrite value.
 - **Type 2:** Insert new row with surrogate key + version.
 - **Type 3:** Add new column.
👉 *Implemented with:* Databricks Delta Merge or ADF Mapping Data Flows.
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4. Explain Delta Lake and its advantages over traditional data lakes.

- Adds **ACID transactions**, schema enforcement, time travel, merges.
 - Solves small files problem with compaction.
👉 Ensures reliability in Data Lake pipelines.
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5. How do you optimize a PySpark job for performance?

- Repartition data smartly.
 - Use broadcast joins for small tables.
 - Cache reused DataFrames.
 - Enable AQE (Adaptive Query Execution).
 - 👉 Always monitor Spark UI for bottlenecks.
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6. What are the best practices for partitioning data in Spark?

- Partition by frequently filtered columns.
 - Avoid high-cardinality columns.
 - Optimize partition size (~128 MB).
 - 👉 *Example:* Partition by order_date for time-series queries.
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7. How do you handle schema evolution in a data pipeline?

- Use formats like Delta/Parquet.
 - Auto-merge schema in Delta.
 - Add default values for new columns.
 - 👉 Ensures backward compatibility in ETL.
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8. Explain the concept of a Data Mesh and its benefits.

- **Data Mesh:** Decentralized architecture → data owned by domain teams.
 - Promotes **self-service**, scalability, domain responsibility.
 - 👉 Large enterprises adopt this for distributed teams.
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9. How do you choose the right file format for a big data pipeline (Parquet vs ORC vs Avro)?

- **Parquet:** Columnar, good for analytics.
- **ORC:** Optimized for Hive queries.
- **Avro:** Row-based, good for streaming.
- 👉 *Best practice:* Use Parquet/Delta for analytical workloads.

10. Describe your approach to designing a star schema.

- Identify **fact table** (measurable metrics).
- Identify **dimension tables** (attributes).
- Define surrogate keys.
👉 *Example:* Sales fact with dimensions like Customer, Product, Time.

11. Explain Azure Data Factory integration with Azure Databricks.

- ADF orchestrates → Databricks runs transformations.
- ADF triggers Databricks notebooks via activities.
👉 *Example:* ADF pipeline calls Databricks job for data cleansing.

12. How do you ensure data quality and validation in ETL workflows?

- Apply row count checks.
- Validate schema.
- Use hash totals, duplicates removal.
👉 Store failed records in error tables for analysis.

13. What is the purpose of Z-ordering in Databricks Delta tables?

- Multi-dimensional clustering to speed queries.
- Groups related rows together.
👉 *Example:* Speeding queries filtering on customer_id.

14. Explain how caching works in Spark and when to use it.

- Stores DataFrame in memory for reuse.
- Use when multiple actions performed on same dataset.
👉 *Example:* Cache lookup tables reused across jobs.

15. How do you handle real-time streaming data in Azure or AWS?

- Azure: Event Hub + Databricks Structured Streaming.
 - AWS: Kinesis + Glue/Spark.
👉 *Example:* Real-time fraud detection from transaction stream.
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16. What are best practices for securing data pipelines in the cloud?

- Use RBAC & Managed Identity.
 - Encrypt at rest & transit.
 - Store secrets in Key Vault.
👉 Ensure compliance with GDPR/ISO.
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17. How would you approach building a data catalog?

- Use tools like **Azure Purview** or **Collibra**.
 - Document schema, lineage, owners.
👉 Improves discoverability & governance.
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18. Explain Unity Catalog in Databricks.

- Centralized governance & fine-grained access control.
 - Supports auditing, lineage.
👉 Enables consistent security across all workspaces.
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19. How do you monitor and troubleshoot Spark job failures?

- Use Spark UI to identify skew/shuffle issues.
 - Check logs in Databricks.
 - Apply retries, fix partitioning.
👉 *Example:* Debug slow stage due to skewed join.
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20. What are the different cluster modes in Spark?

- **Local Mode:** Single machine.
- **Standalone Mode:** Spark's own cluster manager.

- **YARN/Kubernetes:** For distributed resource management.
👉 Cloud Databricks = managed cluster mode.
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21. How do you design a high-availability data pipeline?

- Use multiple regions (geo-redundancy).
 - Enable retries, failover.
 - Stateless components with checkpoints.
👉 Example: Streaming pipeline with checkpoint restart.
-

22. What is the difference between OLTP and OLAP systems?

- **OLTP:** Transactional, real-time inserts/updates (e.g., SQL Server).
 - **OLAP:** Analytical, read-heavy, aggregates (e.g., Synapse).
👉 OLTP for banking apps, OLAP for dashboards.
-

23. How do you handle late-arriving data in ETL?

- Use watermarking (streaming).
 - Merge into Delta tables.
👉 Example: Late order events arriving next day processed via Delta Merge.
-

24. Explain how orchestration works in ADF pipelines.

- Pipelines sequence multiple activities.
 - Use dependencies, triggers, parameters.
👉 Example: Ingest → Transform → Notify sequence.
-

25. What are best practices for partitioning and bucketing in Spark?

- Partition on high-cardinality filters (e.g., date).
 - Bucket on join keys for shuffle optimization.
👉 Example: Bucket customer_id to improve joins.
-

26. How do you handle large data migrations to cloud data warehouses?

- Use staging (Blob/ADLS).
 - Use PolyBase/Copy activity.
 - Parallel load with batch partitioning.
👉 Example: 5 TB Oracle → Synapse migration.
-

27. Explain row-level and column-level security in Azure Synapse.

- **Row-level:** Restrict rows (e.g., region-wise access).
 - **Column-level:** Mask/hide sensitive columns.
👉 Example: Sales manager sees only their region's data.
-

28. How do you implement CI/CD for data pipelines?

- Use Azure DevOps/GitHub Actions.
 - Store pipelines in JSON.
 - Deploy via ARM templates or APIs.
👉 Example: Deploy ADF pipelines via DevOps release pipeline.
-

29. What is the difference between Lambda and Kappa architecture?

- **Lambda:** Separate batch + streaming pipelines.
 - **Kappa:** Streaming-only, batch handled as replay.
👉 Kappa reduces complexity, but Lambda gives flexibility.
-

30. How do you approach cost optimization in cloud data engineering?

- Auto-terminate clusters.
- Archive cold data.
- Use serverless/on-demand instead of always-on compute.
👉 Example: Serverless Synapse for ad-hoc queries instead of dedicated pool.

1. Explain your experience with data pipeline orchestration tools.

- Used **Azure Data Factory (ADF)**, **Apache Airflow**, and **Databricks Jobs**.
 - Orchestrated ingestion (SQL, APIs), transformations (PySpark), and loads (Synapse/Delta).
👉 *Example:* Automated daily sales ETL with ADF scheduling + Databricks transformations.
-

2. How would you design a batch vs real-time pipeline?

- **Batch:** Scheduled jobs (daily/hourly) using ADF + Databricks.
 - **Real-time:** Event-driven using Event Hub/Kafka + Spark Structured Streaming.
👉 Batch for reports, real-time for fraud detection.
-

3. What are Delta Live Tables and their benefits?

- Managed ETL framework in Databricks.
 - Declarative pipeline definition with **auto-dependencies, quality checks, and schema enforcement**.
👉 Benefit: Less manual coding, more reliability.
-

4. How do you handle incremental loads in ETL?

- Capture only changes (CDC).
 - Use watermarks or last_modified_date.
 - Apply MERGE in Delta or Synapse.
👉 *Example:* Load only yesterday's modified rows, not full table.
-

5. What is the difference between managed and external tables in Hive or Databricks?

- **Managed:** Metadata + data managed by system. Dropping deletes both.
 - **External:** Metadata only, data remains in storage.
👉 External used when multiple tools need to access same data.
-

6. Explain the concept of schema-on-read vs schema-on-write.

- **Schema-on-read:** Flexible, applied at query time (Data Lake).
 - **Schema-on-write:** Enforced at load time (Warehouse).
👉 Example: ADLS JSON logs (read schema later) vs Synapse fact tables (schema enforced upfront).
-

7. How do you debug slow-running Spark jobs?

- Check Spark UI for bottlenecks.
 - Optimize joins, repartition, avoid shuffles.
 - Enable AQE, cache frequently reused data.
👉 Example: Found skewed join → applied salting fix.
-

8. What are different join strategies in Spark?

- Shuffle hash join.
 - Broadcast join.
 - Sort merge join.
 - Shuffle nested loop join.
👉 Choose based on data size and skew.
-

9. How do you implement error handling in ETL workflows?

- ADF: Use **On Failure** activities, send alerts.
 - Databricks: Try/Except with logging.
 - Store failed records in error tables for reprocessing.
👉 Example: Send email alert + log failed file path in SQL table.
-

10. What is the difference between ADF Copy Activity and Data Flows?

- **Copy Activity:** EL (move/load data).
- **Data Flows:** ETL (transformations at scale).
👉 Copy for raw ingestion, Data Flows for cleansing & transformations.

11. How do you design data quality frameworks for pipelines?

- Define rules (row count, schema, duplicates, null checks).
- Apply in Databricks/ADF Data Flows.
- Store results in audit tables.
👉 Example: Null check on customer_id → reject bad records.

12. Explain how to use PySpark Window functions with examples.

- Enable advanced analytics like ranking, running totals.
- Example: ROW_NUMBER() OVER (PARTITION BY customer ORDER BY date) → get latest order per customer.
👉 Useful for deduplication, trend analysis.

13. What are Bloom filters in Delta Lake?

- Probabilistic index that speeds up searches on large datasets.
- Reduces scan time significantly.
👉 Example: Filter customers by email faster with Bloom filters.

14. How do you ensure GDPR compliance in data engineering projects?

- Mask/Encrypt PII data.
- Implement data retention policies.
- Audit data lineage.
👉 Example: Encrypt SSN column in ADLS, track access via Purview.

15. What is an event-driven architecture in data engineering?

- System reacts to **events** instead of schedule.
 - Events trigger pipelines (file drop, message in Event Hub).
👉 Example: New file in Blob triggers ingestion pipeline.
-

16. Explain the concept of data shuffling in Spark.

- Redistribution of data across partitions.
 - Happens in joins, groupBy, distinct.
 - Expensive due to network + disk I/O.
👉 Minimize shuffles with partitioning and broadcast joins.
-

17. How do you schedule jobs in Databricks?

- Use **Databricks Job Scheduler** or trigger from ADF.
 - Supports retries, notifications, cron scheduling.
👉 Example: Daily ETL notebook scheduled in Databricks Jobs.
-

18. Explain checkpointing in streaming pipelines.

- Saves metadata (offsets, progress) in persistent storage.
 - Allows restart from last checkpoint on failure.
👉 Example: Kafka → Spark streaming with checkpoint in ADLS.
-

19. How do you optimize cluster sizing for big data jobs?

- Right-size based on dataset size & workload.
 - Use auto-scaling clusters.
 - Avoid over-provisioning → increases cost.
👉 Example: Large batch → 8-node cluster, small jobs → 2-node.
-

20. How do you automate schema validation in data pipelines?

- Compare source vs target schema.
 - Use automated schema validation scripts in Databricks.
 - Store schema versions in metadata tables.
👉 Example: Reject file if new column not expected.
-

21. How do you choose between Synapse, Redshift, and BigQuery?

- **Synapse:** Best with Azure ecosystem.
 - **Redshift:** AWS native.
 - **BigQuery:** Serverless, good for ad-hoc queries.
👉 Choice depends on cloud strategy & use case.
-

22. Explain how Databricks Autoloader works.

- Incrementally loads new files from cloud storage.
 - Uses file notification or directory listing.
 - Handles schema evolution automatically.
👉 Example: Load new daily CSVs into Delta table without manual intervention.
-

23. How do you manage secrets in ADF or Databricks?

- Store in **Azure Key Vault**.
 - ADF/Databricks can reference secrets dynamically.
👉 Example: SQL password stored in Key Vault, retrieved at runtime.
-

24. Explain a scenario where you used partition pruning.

- Partition pruning reads only required partitions.
 - Example: Sales data partitioned by year. Querying 2023 reads only that partition, not full dataset.
👉 Improves query performance & cost.
-

25. How do you optimize S3 or ADLS storage for analytics workloads?

- Use columnar formats (Parquet, Delta).
 - Avoid small files → compact data.
 - Organize with partitions.
👉 Example: Compact 10,000 small files into fewer Parquet files.
-

26. How do you migrate ETL pipelines from on-premises to cloud?

- Assess current pipelines.
 - Rebuild in ADF/Databricks.
 - Use Lift-and-Shift (SSIS IR in ADF) if needed.
👉 Example: Migrate Oracle ETL to ADF with self-hosted IR.
-

27. Explain row vs columnar storage.

- **Row storage:** OLTP, fast inserts/updates.
 - **Columnar storage:** OLAP, fast aggregation/analytics.
👉 Example: SQL Server (row), Parquet/ORC (columnar).
-

28. How do you debug failures in ADF pipeline runs?

- Check ADF Monitor hub for activity logs.
 - View integration runtime logs.
 - Enable diagnostic logs in Log Analytics.
👉 Example: Found failure due to missing Linked Service credentials.
-

29. How do you use ADF triggers effectively?

- Types: **Schedule, Tumbling Window, Event-based, Manual.**
 - Event trigger → fires on file arrival.
👉 Example: Event trigger starts pipeline when file lands in ADLS.
-

30. What is your approach to handling schema drift in data pipelines?

- Enable schema drift in Data Flows.
 - Store unknown columns in staging.
 - Validate schema versions.
👉 Ensures pipelines don't fail when extra columns appear.
-

Wipro – Data Engineer Interview Questions & Answers

1. Explain the role of Delta Lake in modern data pipelines.

- Delta Lake is a **storage layer** on top of Data Lake that provides **ACID transactions**, schema enforcement, and time travel.
 - Solves issues like small files, unreliable queries, and schema evolution.
👉 *Example:* Replace daily full refresh with incremental **MERGE** into Delta tables.
-

2. How do you create a Slowly Changing Dimension Type 2 pipeline?

- Track history by inserting a new row with surrogate key and validity dates.
 - Implemented using **MERGE** in Delta Lake or ADF Data Flows.
👉 *Example:* Customer changed address → old record closed, new record added with new validity dates.
-

3. How do you handle large data ingestion from APIs?

- Use **pagination & partitioning** to break requests.
 - Use **ADF REST connector** or **Databricks REST APIs**.
 - Implement retry logic for API failures.
👉 *Example:* Ingest millions of product records from e-commerce API in batches.
-

4. Explain how Spark Catalyst optimizer works.

- Catalyst Optimizer is Spark's query optimizer.
 - Optimizes logical plan → physical plan → executes efficiently.
 - Uses predicate pushdown, constant folding, projection pruning.
👉 *Example:* SELECT * WHERE region='US' → pushes filter early to avoid scanning entire dataset.
-

5. What is Adaptive Query Execution (AQE) in Spark?

- AQE dynamically optimizes queries at runtime.
 - Fixes skew, adjusts number of reducers, selects better join strategy.
👉 *Example:* Large skewed joins automatically switched to broadcast join.
-

6. How do you handle skewed data in joins?

- Add **salting** to distribute skewed keys.
 - Use broadcast joins.
 - Repartition data by balanced keys.
👉 *Example:* One popular product generates 80% of sales rows → salt key to balance load.
-

7. What is the difference between DataFrame and Dataset API?

- **DataFrame:** Untyped, schema-aware, optimized with Catalyst.
 - **Dataset:** Typed API (in Scala/Java), combines benefits of RDDs + DataFrames.
👉 PySpark uses DataFrames only (no Dataset API).
-

8. Explain the concept of broadcast join.

- Small dataset is broadcasted to all worker nodes → avoids shuffle.
 - Faster when one table is small enough (<100 MB).
👉 *Example:* Sales fact (huge) joined with country lookup (small).
-

9. How do you handle late-arriving data in a streaming pipeline?

- Use **watermarking** to allow late events within a time window.
 - Merge into Delta table with upserts.
👉 *Example:* Event arrives 5 mins late → still included within watermark of 10 mins.
-

10. How do you secure ADF pipelines?

- Use Managed Identity for authentication.
 - Store secrets in Key Vault.
 - Restrict access with RBAC & private endpoints.
👉 *Example:* Developer cannot see SQL password → retrieved securely at runtime.
-

11. What are PolyBase and its use cases in Synapse?

- PolyBase enables querying external data (ADLS/Blob/Hadoop) using T-SQL.
 - Use for fast parallel loads into Synapse.
👉 *Example:* Load 1 TB parquet file from ADLS directly into Synapse fact table.
-

12. How do you use Databricks for ML feature engineering?

- Clean raw data, create aggregates, window functions, embeddings.
 - Store features in Delta/Feature Store.
👉 *Example:* Generate customer churn features (purchase frequency, avg spend).
-

13. Explain Medallion Architecture in Databricks.

- **Bronze:** Raw ingestion.
 - **Silver:** Cleansed data.
 - **Gold:** Aggregated/curated data.
👉 Industry standard for scalable pipelines.
-

14. How do you automate pipeline deployments?

- Store pipelines/notebooks in Git.
 - Use CI/CD tools (Azure DevOps, GitHub Actions).
 - Deploy via ARM templates/APIs.
👉 *Example:* Deploy ADF pipelines automatically to UAT/Prod.
-

15. How do you perform incremental load using watermarking?

- Use watermark column (timestamp).
 - Load only records newer than last processed timestamp.
👉 *Example:* Load only yesterday's new orders, not full history.
-

16. How do you set up monitoring for ETL pipelines?

- ADF: Use Monitor Hub + Log Analytics.
 - Databricks: Use job logs + metrics dashboards.
 - Configure alerts for failures.
👉 *Example:* Send Teams/Slack notification on failure.
-

17. How do you handle out-of-order events in streaming systems?

- Use watermark + event-time windowing.
 - Buffer events until watermark threshold.
👉 *Example:* Events from IoT device processed in correct time sequence despite delays.
-

18. Explain how Azure Event Hub integrates with Spark Streaming.

- Event Hub = ingestion service, Spark reads events in real-time.
 - Use Spark connector for Event Hub.
👉 *Example:* Streaming tweets from Event Hub into Spark → sentiment analysis.
-

19. How do you implement retry policies in ADF?

- Configure retries at activity level.
 - Use exponential backoff.
👉 *Example:* Retry API ingestion 3 times before failing pipeline.
-

20. Explain partition discovery in Spark.

- Spark automatically identifies partitioned columns in file paths.
 - Uses folder structure (/year=2023/month=08) for pruning.
👉 *Example:* Query with WHERE year=2023 → only that partition scanned.
-

21. How do you handle duplicates in data pipelines?

- Use Spark dropDuplicates().
- Use window functions with row_number().
👉 *Example:* Keep latest customer record based on timestamp.

22. Explain the concept of snapshot isolation in Delta tables.

- Queries see a consistent snapshot even when updates are happening.
- Prevents dirty reads, ensures ACID compliance.
👉 *Example:* Analysts see stable version while ETL is merging new data.

23. How do you design data vault modeling?

- Consists of **Hubs** (business keys), **Links** (relationships), and **Satellites** (descriptive data).
- Supports auditability and scalability.
👉 *Example:* Customer Hub + Order Hub linked via Order-Customer Link.

24. How do you manage pipeline dependencies?

- Use **Execute Pipeline** activity in ADF.
- Apply triggers with dependencies.
- In Databricks, use job workflows with task dependencies.
👉 *Example:* Run transformation pipeline only after ingestion finishes.

25. What is the difference between CDC and full refresh ETL?

- **CDC:** Loads only changed records (faster, cheaper).
- **Full refresh:** Reloads entire dataset (slower, costly).
👉 Use CDC for daily loads, full refresh for one-time initial loads.

26. How do you choose between Azure Data Explorer and Synapse?

- **Azure Data Explorer (ADX):** For time-series/log data, near real-time queries.
- **Synapse:** For structured, BI-focused workloads.
👉 *Example:* ADX for telemetry analysis, Synapse for sales reports.

27. How do you troubleshoot performance issues in ADF?

- Check integration runtime performance.
 - Optimize batch size and degree of parallelism.
 - Use PolyBase for large loads.
👉 Example: Optimized Copy activity by splitting files into partitions.
-

28. Explain concurrency scaling in data warehouses.

- Allows multiple queries to run concurrently without bottlenecks.
 - Synapse/Redshift add temporary compute nodes for concurrency.
👉 Example: 100 users querying simultaneously without performance degradation.
-

29. How do you handle multi-cloud data movement?

- Use cross-cloud connectors (ADF, Informatica).
 - Store in common format (Parquet/Delta).
 - Apply secure transfer (TLS, encryption).
👉 Example: Move AWS S3 data into Azure ADLS for analytics.
-

30. What is the purpose of compaction in Delta tables?

- Small file problem → too many tiny files slow queries.
 - Compaction merges small files into larger files.
👉 Example: Compact 10,000 small JSON files into optimized Parquet files.
-

TCS – Data Engineer Interview Questions & Answers

1. Explain the architecture of a typical data pipeline you built.

- **Ingestion:** ADF ingests raw data (CSV, SQL, APIs) into ADLS.
- **Processing:** Databricks (PySpark) cleanses and transforms.
- **Storage:** Delta Lake (Bronze, Silver, Gold).
- **Consumption:** Synapse/Power BI for analytics.
👉 Example: End-to-end retail sales pipeline with near real-time reporting.

2. What are the differences between OLTP and OLAP systems?

- **OLTP:** Transactional, fast inserts/updates, row storage (e.g., SQL Server).
- **OLAP:** Analytical, read-heavy, aggregates, columnar storage (e.g., Synapse).
👉 OLTP = order booking, OLAP = sales trends dashboard.

3. How do you implement Change Data Capture (CDC)?

- Capture only inserts/updates/deletes from source.
- Use **Debezium, ADF Incremental Copy, Delta MERGE**.
👉 Example: Oracle → ADF incremental load → Delta Silver table.

4. Explain different Spark join types.

- Inner, Left/Right, Full Outer, Semi, Anti, Cross Join.
- Optimizations: Broadcast join, repartition.
👉 Example: Fact table + dimension table join using broadcast.

5. How do you handle schema evolution in Delta Lake?

- Enable mergeSchema in write.
- ALTER TABLE to add/remove columns.
👉 Example: Adding email_id to customer table without dropping history.

6. What is the difference between partitioning and bucketing?

- **Partitioning:** Splits data by column values.
- **Bucketing:** Hash distributes rows into fixed buckets.
👉 Partition on year, bucket on customer_id for joins.

7. Explain Data Vault modeling.

- Hubs (business keys), Links (relationships), Satellites (descriptive).

- Scalable, auditable, flexible for big data.
👉 Example: Customer hub + order hub linked with order-customer link.
-

8. How do you optimize Spark shuffle operations?

- Repartition wisely.
 - Use broadcast joins.
 - Enable AQE.
👉 Example: Avoid unnecessary groupBy that causes shuffle.
-

9. What are the advantages of Parquet over CSV?

- Columnar storage → faster queries.
 - Compression built-in → less storage cost.
 - Schema-aware.
👉 Example: Store sales fact table in Parquet for BI.
-

10. How do you manage pipeline failures?

- Retry logic in ADF/Databricks.
 - Store error records separately.
 - Send alerts via email/Teams.
👉 Example: Failed file path logged in SQL error table.
-

11. Explain role of PolyBase in Synapse.

- Allows external data query (ADLS, Blob, Hadoop).
 - High performance parallel loads.
👉 Example: Query raw parquet files directly in Synapse.
-

12. How do you secure sensitive data (PII)?

- Encrypt at rest & transit.
- Mask data in Synapse.

- Use Key Vault for secrets.
👉 Example: Encrypt Aadhaar numbers in ADLS.
-

13. What is the Medallion Architecture?

- **Bronze:** Raw ingestion.
 - **Silver:** Clean, structured.
 - **Gold:** Aggregated, business-ready.
👉 Standard for Lakehouse in Databricks.
-

14. How do you design incremental pipelines in ADF?

- Use watermark column.
 - Store last load timestamp in config table.
 - Load only new/updated rows.
👉 Example: Load only new orders > last load date.
-

15. Explain Spark DAG execution.

- DAG = Directed Acyclic Graph of transformations.
 - Spark builds DAG → splits into stages → tasks executed.
👉 Lazy evaluation: execution triggered only by action.
-

16. How do you monitor data pipelines in production?

- ADF: Monitor Hub, Log Analytics.
 - Databricks: Job UI, logs.
 - Alerts to Ops team.
👉 Example: Pipeline failure triggers Teams alert.
-

17. What are Bloom filters in Delta Lake?

- Index for faster search in large datasets.

- Reduces full scan cost.
👉 Example: Query by email_id much faster.
-

18. How do you implement surrogate keys in data warehouse?

- Auto-generated keys (IDENTITY/UUID).
 - Avoids using natural keys which can change.
👉 Example: Customer surrogate key used instead of SSN.
-

19. What is Event Hub, and how does it integrate with Databricks?

- Event Hub = real-time data ingestion service.
 - Databricks Structured Streaming consumes events.
👉 Example: Streaming IoT device logs into Delta tables.
-

20. How do you optimize Synapse queries?

- Use partitioned tables.
 - Distribute on join keys.
 - Use result set caching.
👉 Example: Distribute sales fact on customer_id to improve joins.
-

21. Explain Time Travel in Delta tables.

- Query older versions of data using version or timestamp.
👉 Example: VERSION AS OF 10 → rollback to past data for auditing.
-

22. How do you use ADF triggers?

- Schedule: Time-based.
 - Tumbling window: Batch periodic.
 - Event: On file arrival.
👉 Example: Event trigger runs pipeline when file lands in ADLS.
-

23. Explain different types of SCD.

- Type 1: Overwrite.
 - Type 2: Insert new row, track history.
 - Type 3: Add new column.
👉 Example: Track customer address changes with Type 2.
-

24. How do you reduce cloud costs in data pipelines?

- Use auto-terminate clusters.
 - Compact files.
 - Use serverless compute.
👉 Example: Archive 1-year-old logs to cold storage.
-

25. How do you test data pipelines?

- Unit test transformations.
 - Validate row counts, schema.
 - Regression test with known inputs/outputs.
👉 Example: Ensure sum(sales) in target = source.
-

26. What are common Spark optimization techniques?

- Use DataFrames, not RDDs.
 - Cache intermediate results.
 - Optimize shuffle.
👉 Example: Broadcast lookup tables.
-

27. Explain row-level security in Synapse.

- Restricts row access based on user.
👉 Example: Regional manager sees only their region's sales.
-

28. How do you orchestrate multiple pipelines?

- ADF Execute Pipeline activity.
 - Dependencies with success/failure conditions.
👉 Example: Transformation runs after ingestion completes.
-

29. What is Unity Catalog?

- Centralized governance in Databricks.
 - Manages permissions, lineage, audit.
👉 Example: Control access to Delta tables across workspaces.
-

30. How do you manage schema drift?

- Enable schema drift in ADF Data Flows.
 - Store unknown columns separately.
 - Version schema in metadata.
👉 Example: New column in source captured without breaking pipeline.
-

Virtusa – Data Engineer Interview Questions & Answers

1. Explain Data Lakehouse architecture.

- Combines benefits of **Data Lakes (scalable, raw storage)** and **Data Warehouses (structured, ACID, BI-ready)**.
 - Powered by technologies like **Databricks Delta Lake**.
👉 Example: Store raw IoT data (lake) + curated sales analytics (warehouse) in one system.
-

2. How do you perform CDC (Change Data Capture) in Databricks?

- Use **Delta Lake MERGE** to apply inserts/updates/deletes.
 - Source CDC logs from SQL, Kafka, or ADF incremental copy.
👉 Example: Daily Oracle table changes merged into Delta Silver table.
-

3. What is the difference between structured, semi-structured, and unstructured data?

- **Structured:** Tabular (SQL tables).
 - **Semi-structured:** JSON, XML, Parquet.
 - **Unstructured:** Images, videos, audio, logs.
👉 Example: Bank transactions (structured), sensor JSON (semi), CCTV footage (unstructured).
-

4. How do you implement a Data Quality Framework?

- Define rules (null check, duplicates, schema validation).
 - Automate checks in ADF or PySpark.
 - Store results in audit logs.
👉 Example: Reject records with missing customer_id.
-

5. Explain difference between Data Engineer vs Data Scientist.

- **Data Engineer:** Focuses on pipelines, ETL, infrastructure.
 - **Data Scientist:** Focuses on ML models, predictions.
👉 They collaborate: Engineer preps data → Scientist builds models.
-

6. How do you handle schema evolution in Parquet/Delta?

- Enable mergeSchema in Delta.
 - Maintain version history of schema.
👉 Example: Add email column to customer table without reprocessing.
-

7. What is Z-ordering in Databricks?

- Optimizes multi-dimensional queries.
 - Co-locates related rows for faster filtering.
👉 Example: Speed up queries filtering on customer_id and region.
-

8. How do you optimize PySpark jobs for performance?

- Repartition wisely.

- Use broadcast joins.
 - Cache reused data.
 - Enable AQE.
👉 Example: Broadcast small dimension table to speed joins.
-

9. What are different cluster modes in Spark?

- Local mode, Standalone, YARN, Kubernetes.
👉 Databricks = managed cluster mode on cloud.
-

10. Explain Medallion Architecture.

- Bronze = Raw.
 - Silver = Cleaned.
 - Gold = Curated/aggregated.
👉 Industry standard for data pipelines.
-

11. How do you design star schema for analytics?

- Central fact table with measures.
 - Dimension tables with descriptive attributes.
👉 Example: Sales fact + Customer, Product, Time dimensions.
-

12. How do you secure pipelines in Azure?

- Use Managed Identity.
 - Secrets in Key Vault.
 - RBAC + network restrictions.
👉 Example: ADF connects to SQL via Key Vault secrets.
-

13. Explain Bloom filters.

- Probabilistic index for fast search.

- Used in Delta for faster filtering.
👉 Example: Customer lookup by email much faster.
-

14. How do you manage pipeline failures?

- Use retries.
 - Log failed records.
 - Alerts via email/Teams.
👉 Example: Error table logs failed customer records.
-

15. What is Adaptive Query Execution in Spark?

- Dynamically optimizes queries at runtime.
 - Fixes skew, adjusts reducers, changes join strategy.
👉 Example: Automatically switches to broadcast join.
-

16. How do you implement CI/CD for ADF and Databricks?

- Store JSON/notebooks in Git.
 - Use Azure DevOps pipelines.
 - Deploy via ARM templates/APIs.
👉 Example: Deploy ADF pipelines from Dev → Prod.
-

17. Explain Data Skew. How to handle it?

- Uneven data distribution across partitions.
 - Fix with salting, repartitioning, broadcast joins.
👉 Example: One key dominates → add random salt.
-

18. How do you optimize Synapse queries?

- Partition big tables.
- Distribute on join keys.

- Use materialized views.
👉 Example: Partition sales fact on order_date.
-

19. What is the difference between Lakehouse and Data Warehouse?

- **Warehouse:** Structured, schema-on-write.
 - **Lakehouse:** Combines raw + structured, ACID with Delta.
👉 Lakehouse is modern evolution.
-

20. Explain Unity Catalog.

- Centralized governance for Databricks.
 - Fine-grained permissions, lineage, audit.
👉 Example: Restrict access to sensitive tables.
-

21. How do you monitor ADF pipelines?

- Use Monitor Hub.
 - Enable diagnostics with Log Analytics.
 - Configure alerts.
👉 Example: Failed pipeline sends Teams alert.
-

22. Explain Window functions in PySpark.

- Perform calculations across partitions.
 - Examples: Ranking, running totals, lag/lead.
👉 Example: Find latest transaction per customer.
-

23. What are best practices for handling large files in Data Lake?

- Use columnar formats.
- Compact small files.
- Partition data by query filters.
👉 Example: Compact JSON into Parquet.

24. How do you handle late-arriving data in ETL?

- Use watermarking in streaming.
- Merge in Delta Lake.
👉 Example: Late sales order merged next day.

25. What are common performance tuning techniques in Spark?

- Avoid shuffles.
- Use broadcast joins.
- Persist intermediate results.
👉 Example: Cache dimension table reused in multiple joins.

26. Explain difference between Lambda vs Kappa architecture.

- **Lambda:** Separate batch + streaming.
- **Kappa:** Streaming-only (batch = replay).
👉 Lambda = flexible, Kappa = simpler.

27. How do you validate ETL outputs?

- Row counts.
- Hash totals.
- Business rule validations.
👉 Example: Total sales = source records.

28. Explain columnar storage benefits.

- Stores by column, not row.
- Better compression, faster aggregates.
👉 Example: Parquet faster than CSV for BI queries.

29. How do you handle schema drift in ADF?

- Enable schema drift in Data Flows.
 - Store unexpected columns separately.
👉 Example: New column appears in source → pipeline still runs.
-

30. How do you optimize cloud costs in Databricks?

- Auto-terminate clusters.
- Use spot instances.
- Compact small files.
👉 Example: Archive old data to cold storage tier.

Coforge – Data Engineer Interview Questions & Answers

1. What is the difference between ETL and ELT?

- **ETL:** Extract → Transform → Load (transform before loading). Traditional DWH approach.
 - **ELT:** Extract → Load → Transform (transform after loading into target). Modern cloud approach.
👉 Example: ELT in Synapse with raw data loaded first, transformations done via SQL.
-

2. How do you design a data pipeline for near real-time analytics?

- Use Event Hub/Kafka for ingestion.
 - Process with Spark Structured Streaming.
 - Store in Delta Lake or Synapse.
👉 Example: Real-time fraud detection system.
-

3. What is the benefit of Delta Lake over plain Parquet?

- Delta provides **ACID transactions**, schema enforcement, time travel, and upserts.
👉 Example: Handle late-arriving events with Delta MERGE.
-

4. How do you optimize Spark partitions?

- Ideal partition size ~128 MB.
 - Use repartition() for large reshuffles.
 - Use coalesce() to reduce partitions without shuffle.
👉 Example: Reduce 500 tiny partitions into 20 larger ones.
-

5. What is data lineage and why is it important?

- Tracks flow of data from source → transformation → destination.
 - Helps with debugging, auditing, compliance.
👉 Example: Using Azure Purview to trace sales data origin.
-

6. Explain the concept of surrogate keys in data warehouses.

- Artificial keys (auto-generated) instead of natural business keys.
 - Prevents dependency on changing business keys.
👉 Example: Customer surrogate key instead of SSN.
-

7. How do you implement incremental data loading in ADF?

- Use watermark columns (last modified).
 - Store last processed value in config.
👉 Example: Load only rows updated after last run.
-

8. Explain star schema vs snowflake schema.

- **Star:** Simple, fact connected directly to dimensions.
 - **Snowflake:** Normalized dimensions.
👉 Example: Sales fact with dimensions (Customer, Product, Time).
-

9. How do you handle slowly changing dimensions (SCD Type 2)?

- Insert new row with surrogate key and validity dates.

- Maintain history of changes.
👉 Example: Customer address changes tracked with old + new records.
-

10. What are PolyBase external tables in Synapse?

- Allow querying external files (ADLS/Blob).
 - Used for high-performance data loads.
👉 Example: Query parquet files directly without ingestion.
-

11. How do you manage secrets securely in cloud pipelines?

- Store in Key Vault.
 - Access with Managed Identity.
👉 Example: ADF retrieves SQL credentials from Key Vault.
-

12. What is Z-ordering and why use it?

- Optimizes multi-dimensional queries.
 - Co-locates related rows.
👉 Example: Query filtering on region + customer_id is faster.
-

13. How do you handle duplicate records in pipelines?

- Use Spark dropDuplicates().
 - Use window functions with row_number.
👉 Example: Deduplicate customer transactions.
-

14. Explain Medallion architecture.

- Bronze = Raw, Silver = Cleansed, Gold = Aggregated.
👉 Example: IoT raw → cleaned → analytics-ready in Gold.
-

15. How do you monitor Databricks jobs?

- Use job UI & cluster logs.

- Send logs to Log Analytics.
👉 Example: Alerts configured for failed jobs.
-

16. What is the difference between batch and streaming data processing?

- **Batch:** Large chunks, higher latency.
 - **Streaming:** Continuous, near real-time.
👉 Batch = monthly reports, Streaming = live dashboards.
-

17. How do you optimize join performance in Spark?

- Use broadcast joins for small tables.
 - Repartition on join keys.
 - Handle skew with salting.
👉 Example: Broadcast country lookup in sales join.
-

18. How do you implement pipeline orchestration in ADF?

- Pipelines link multiple activities.
 - Use triggers, dependencies, and Execute Pipeline.
👉 Example: Ingestion → Transformation → Load sequence.
-

19. What is Adaptive Query Execution (AQE) in Spark?

- Dynamically adjusts join strategy & partitions at runtime.
👉 Example: Automatically switch to broadcast join.
-

20. Explain Unity Catalog in Databricks.

- Provides centralized data governance.
 - Fine-grained access, lineage, and auditing.
👉 Example: Enforce role-based access to Delta tables.
-

21. How do you troubleshoot ADF pipeline failures?

- Check Monitor Hub logs.
 - View Integration Runtime logs.
 - Configure diagnostic logs in Log Analytics.
👉 Example: Found missing SQL credentials caused pipeline failure.
-

22. What are the benefits of columnar storage formats?

- Better compression.
 - Faster analytical queries.
👉 Example: Parquet files reduce scan cost vs CSV.
-

23. How do you implement retry logic in pipelines?

- Configure retries in ADF activities.
 - Use exponential backoff for APIs.
👉 Example: Retry API ingestion 3 times before failure.
-

24. What is event-driven data ingestion?

- Pipelines triggered by events (file arrival, message queue).
👉 Example: Blob file drop triggers ADF ingestion.
-

25. How do you manage schema drift?

- Enable schema drift in ADF Data Flows.
 - Store unknown columns separately.
👉 Example: New column in source JSON handled dynamically.
-

26. How do you reduce storage costs in Data Lake?

- Use compressed formats (Parquet, Delta).
 - Archive cold data to cheaper tier.
👉 Example: Move 2-year-old logs to archive storage.
-

27. Explain checkpointing in streaming pipelines.

- Stores progress (offsets, state).
 - Enables recovery after failure.
👉 Example: Kafka → Spark streaming resumes from last checkpoint.
-

28. How do you design a high-availability data pipeline?

- Use multiple regions.
 - Failover clusters.
 - Retry logic.
👉 Example: Duplicate Event Hubs in two regions.
-

29. What is the difference between Data Warehouse and Data Mart?

- **Data Warehouse:** Enterprise-wide data repository.
 - **Data Mart:** Subject-area specific (e.g., Sales).
👉 Example: DWH = company-wide, DM = finance department only.
-

30. How do you validate data after ETL load?

- Row count checks.
 - Hash totals.
 - Business rule checks.
👉 Example: Sales totals match between source and target.
-

Amdocs – Data Engineer Interview Questions & Answers

1. What is a Data Pipeline, and what are its key components?

- A data pipeline moves data from **source** → **processing** → **storage** → **consumption**.
 - Components: Ingestion, Transformation, Storage, Orchestration, Monitoring.
👉 Example: Ingest CDR (Call Detail Records) → transform in Spark → store in Delta → BI reports.
-

2. Explain the difference between batch and streaming data processing.

- **Batch:** Processes large chunks on schedule, higher latency.
 - **Streaming:** Processes continuous flow, low latency.
👉 Example: Batch = monthly billing reports, Streaming = real-time fraud alerts.
-

3. How do you implement CDC (Change Data Capture) in Azure?

- Use ADF incremental load with watermark column.
 - Or use Debezium/Kafka for streaming CDC.
 - Apply MERGE in Delta tables.
👉 Example: Capture updated telecom subscriber details daily.
-

4. What are the advantages of Delta Lake in telecom analytics?

- Provides ACID transactions, schema enforcement, time travel, and upserts.
👉 Example: Correct late-arriving telecom usage records via Delta MERGE.
-

5. How do you optimize Spark jobs for telecom call data (billions of records)?

- Use partitioning (by date, region).
 - Broadcast small lookup tables.
 - Enable AQE.
👉 Example: Partition call logs by month for queries.
-

6. Explain partitioning vs bucketing with an example.

- **Partitioning:** Splits by column values.
 - **Bucketing:** Hash distributes into fixed buckets.
👉 Example: Partition on date, bucket on subscriber_id.
-

7. What is Adaptive Query Execution (AQE) in Spark?

- Runtime optimization: fixes skew, adjusts reducers, changes join strategy.
👉 Example: Skewed telecom data automatically rebalanced by AQE.

8. How do you implement error handling in ADF pipelines?

- Use On Failure activities.
- Store failed records in error tables.
- Send alerts via email/Teams.
👉 Example: Log rejected subscriber records.

9. What is Medallion Architecture?

- **Bronze:** Raw telecom CDRs.
- **Silver:** Cleansed/structured.
- **Gold:** Aggregated insights (ARPU, churn).

10. How do you ensure data quality in pipelines?

- Row counts, null checks, duplicates, referential integrity.
👉 Example: Validate that every call record has valid subscriber_id.

11. What is Z-ordering in Delta tables?

- Co-locates related rows for faster filtering.
👉 Example: Optimize queries on subscriber_id + region.

12. How do you handle late-arriving events in streaming pipelines?

- Use watermarking to allow delays.
- Merge late data into Delta.
👉 Example: Calls processed even if logs arrive 10 minutes late.

13. Explain the difference between OLTP and OLAP.

- **OLTP:** Real-time transactions (billing system).
 - **OLAP:** Analytical queries (reporting system).
-

14. How do you secure sensitive customer data?

- Encrypt at rest & transit.
 - Mask PII in Synapse.
 - RBAC & Key Vault for secrets.
👉 Example: Mask subscriber phone numbers in reports.
-

15. How do you orchestrate complex pipelines in ADF?

- Use dependency chaining, Execute Pipeline activity, triggers.
👉 Example: Ingest → Transform → Load → Notify.
-

16. Explain PolyBase in Synapse.

- Query external data in ADLS/Blob.
 - High-performance parallel loads.
👉 Example: Load raw parquet files directly into telecom warehouse.
-

17. How do you troubleshoot Spark job failures?

- Use Spark UI for DAG/stage analysis.
 - Check logs for OOM/shuffle issues.
 - Optimize joins and repartition.
-

18. What is Unity Catalog in Databricks?

- Centralized governance & access control.
 - Provides lineage and auditing.
👉 Example: Manage access to subscriber churn models.
-

19. How do you optimize Synapse queries?

- Partition and distribute tables.
- Use result caching.

- Materialized views for heavy aggregations.
👉 Example: Partition telecom fact by billing cycle.
-

20. How do you manage schema evolution in Delta?

- Enable auto schema merge.
 - ALTER TABLE for new columns.
👉 Example: Add new column plan_type to subscriber table.
-

21. What are Bloom filters in Delta Lake?

- Probabilistic index for faster lookups.
👉 Example: Speed up email/phone number searches in large subscriber table.
-

22. How do you implement pipeline monitoring?

- ADF Monitor Hub + Log Analytics.
 - Databricks job metrics.
 - Alerts for failures.
-

23. What is data skew and how do you fix it?

- Uneven distribution across partitions.
 - Fix with salting, repartitioning, broadcast joins.
👉 Example: One subscriber generating huge data skew.
-

24. How do you optimize file sizes in Delta Lake?

- Compact small files with OPTIMIZE.
 - Target ~128 MB files.
👉 Example: Compact millions of small telecom event logs.
-

25. How do you implement real-time dashboards?

- Ingest with Event Hub.

- Process in Spark Structured Streaming.
 - Store in Delta/Synapse.
 - Visualize with Power BI.
👉 Example: Live network usage dashboard.
-

26. How do you perform pipeline CI/CD?

- Store JSON/notebooks in Git.
 - Use DevOps pipelines for deployment.
👉 Example: Deploy ADF pipelines automatically from Dev → Prod.
-

27. Explain difference between managed vs external tables in Databricks.

- **Managed:** Data + metadata managed by DBFS. Dropping deletes data.
 - **External:** Metadata only, data stored in ADLS.
👉 External preferred for enterprise data lakes.
-

28. How do you reduce cloud costs in data engineering?

- Auto-terminate clusters.
 - Archive cold data.
 - Use spot instances.
👉 Example: Move 2-year-old call logs to archive storage.
-

29. How do you validate incremental loads?

- Compare source vs target counts.
 - Hash totals for data integrity.
👉 Example: Daily subscriber updates validated with counts.
-

30. How do you design a scalable data pipeline?

- Use modular ingestion & transformation layers.
- Decouple storage (Data Lake) from compute (Spark).

- Auto-scale compute.
👉 Example: Scalable telecom CDR processing pipeline.

Accenture – Data Engineer Interview Questions & Answers

1. Explain your approach to building cloud data pipelines.

- Ingest data from multiple sources (SQL, APIs, flat files).
 - Use **ADF/Glue/Dataflow** for orchestration.
 - Transform in **Databricks/Spark**.
 - Store in **Delta Lake/Synapse/BigQuery**.
👉 Example: Marketing data pipeline – raw → cleaned → BI dashboards.
-

2. What is a Data Lakehouse and why is it important?

- Combines **Data Lake flexibility** (raw storage) + **Data Warehouse reliability** (ACID, BI-ready).
 - Powered by Delta Lake, Iceberg, Hudi.
👉 Example: Store raw IoT + curated sales data in one platform.
-

3. How do you implement incremental data loading?

- Use watermark column (last updated).
 - Maintain metadata table for last load.
 - Apply MERGE in Delta.
👉 Example: Load only new invoices after yesterday's timestamp.
-

4. Explain different types of Slowly Changing Dimensions.

- **Type 1:** Overwrite old value.
 - **Type 2:** Add new record with versioning.
 - **Type 3:** Add new column to track history.
👉 Example: Customer address change tracked using Type 2.
-

5. How do you optimize Spark jobs for large-scale data?

- Repartition data smartly.
 - Broadcast small tables.
 - Use AQE.
 - Cache reused datasets.
- 👉 Example: Optimized telecom CDR Spark jobs using partitioning.
-

6. What are the differences between Data Warehouse and Data Mart?

- **Warehouse:** Enterprise-wide, multiple subject areas.
 - **Mart:** Department-specific (e.g., Sales, Finance).
- 👉 Warehouse = entire company, Mart = HR only.
-

7. What is the Medallion Architecture?

- Bronze: Raw ingestion.
 - Silver: Cleansed/structured.
 - Gold: Aggregated, BI-ready.
- 👉 Used in Databricks Lakehouse.
-

8. How do you handle schema drift in pipelines?

- Enable schema drift in ADF Data Flows.
 - Store unknown columns in staging.
 - Validate schema with metadata.
- 👉 Example: Handle new JSON fields without failure.
-

9. What is the difference between ETL and ELT?

- **ETL:** Transform before loading (traditional).
 - **ELT:** Load raw data, transform inside warehouse/lakehouse.
- 👉 Example: ELT in Synapse with transformations post-load.
-

10. Explain PolyBase in Synapse.

- Query external data (ADLS/Blob) with T-SQL.
 - Parallel data loading.
👉 Example: Load parquet files directly into Synapse fact table.
-

11. How do you secure ADF pipelines?

- Use Managed Identity for authentication.
 - Store secrets in Key Vault.
 - Apply RBAC & private endpoints.
👉 Example: Restrict access to SQL DB credentials.
-

12. What are Delta Live Tables?

- Declarative ETL pipelines in Databricks.
 - Automatically handle dependencies, schema enforcement, quality checks.
👉 Example: Define Silver → Gold transformations in DLT.
-

13. How do you handle duplicate records in data pipelines?

- Use Spark dropDuplicates().
 - Use row_number() window function to keep latest.
👉 Example: Deduplicate subscriber transactions.
-

14. Explain difference between OLTP and OLAP systems.

- **OLTP**: Real-time transactions, row storage.
 - **OLAP**: Analytical queries, columnar storage.
👉 OLTP = banking app, OLAP = financial dashboards.
-

15. How do you optimize Synapse performance?

- Distribute fact tables on join keys.
- Partition large datasets.

- Use materialized views.
👉 Example: Distribute sales fact by customer_id.
-

16. What is Z-ordering in Delta Lake?

- Multi-dimensional clustering for faster filtering.
👉 Example: Optimize queries filtering on region + product_id.
-

17. How do you manage pipeline dependencies?

- ADF: Execute Pipeline, Wait, If conditions.
 - Databricks: Job task dependencies.
👉 Example: Run transformation only after ingestion success.
-

18. How do you handle late-arriving data?

- Use watermarking in streaming.
 - Merge into Delta tables.
👉 Example: Late sales order merged next day.
-

19. What is Unity Catalog?

- Centralized governance in Databricks.
 - Fine-grained access control, lineage, auditing.
👉 Example: Role-based access to Delta tables.
-

20. How do you validate ETL pipeline results?

- Row counts.
 - Hash totals.
 - Business rule validation.
👉 Example: Validate total revenue matches source system.
-

21. Explain Adaptive Query Execution (AQE).

- Optimizes Spark queries at runtime.
 - Fixes skew, reduces shuffle partitions, changes join strategies.
👉 Example: AQE switches join to broadcast for small tables.
-

22. How do you design a scalable pipeline?

- Use modular design (Ingestion, Transform, Storage).
 - Decouple compute and storage.
 - Enable auto-scaling clusters.
👉 Example: IoT pipeline scaling for millions of events.
-

23. What is data skew and how do you fix it?

- Skew = uneven distribution of data across partitions.
 - Fix with salting, repartitioning, broadcast joins.
👉 Example: One region generating 80% of rows.
-

24. How do you monitor pipelines in production?

- ADF Monitor Hub + Log Analytics.
 - Databricks jobs UI.
 - Alerts for failures.
👉 Example: Teams alert on failure.
-

25. Explain Data Vault modeling.

- **Hubs** = business keys.
 - **Links** = relationships.
 - **Satellites** = descriptive data.
👉 Example: Customer Hub, Order Hub linked with Order-Customer Link.
-

26. How do you choose between Parquet, Avro, and ORC?

- **Parquet**: Columnar, good for analytics.

- **Avro:** Row-based, good for streaming.
 - **ORC:** Optimized for Hive.
👉 Use Parquet/Delta for analytics workloads.
-

27. How do you troubleshoot ADF pipeline failures?

- Check activity logs in Monitor hub.
 - View Integration Runtime logs.
 - Enable diagnostics in Log Analytics.
👉 Example: Failure due to missing linked service credentials.
-

28. What is checkpointing in Spark Streaming?

- Stores progress & offsets in persistent storage.
 - Allows recovery on failure.
👉 Example: Resume Kafka → Spark streaming job from checkpoint.
-

29. How do you reduce costs in Databricks?

- Auto-terminate clusters.
 - Use spot instances.
 - Compact small files.
👉 Example: Move historical data to cold storage.
-

30. How do you implement CI/CD for data pipelines?

- Store pipelines in Git.
- Use Azure DevOps/GitHub Actions.
- Deploy via ARM templates/APIs.
👉 Example: Automated deployment of ADF & Databricks jobs.

HCL – Data Engineer Interview Questions & Answers

1. What is your experience with Azure Data Factory (ADF)?

- Designed pipelines for **data ingestion, transformation, and orchestration**.
 - Integrated with **Databricks, Synapse, and ADLS**.
👉 Example: Automated daily ingestion from Oracle DB → transformed in Databricks → loaded into Synapse.
-

2. How do you handle incremental data loads in ADF?

- Use **watermark column** (last_updated_date).
 - Store last processed value in metadata/config table.
 - Use query = "WHERE last_updated > @pipeline_param" in Copy activity.
👉 Example: Load only new employee records since last run.
-

3. Explain the difference between Linked Service, Dataset, and Pipeline in ADF.

- **Linked Service:** Connection string to data source.
 - **Dataset:** Schema/structure of data.
 - **Pipeline:** Workflow/orchestration of activities.
👉 Example: Linked Service = SQL DB, Dataset = Customer Table, Pipeline = Load job.
-

4. What is the difference between Mapping Data Flow and Copy Activity?

- **Copy Activity:** EL (extract & load).
 - **Mapping Data Flow:** ETL (transformations at scale).
👉 Example: Copy raw files into ADLS → transform in Mapping Data Flow → load into Synapse.
-

5. How do you manage secrets in ADF?

- Use **Azure Key Vault** integration.
 - Assign Managed Identity to ADF.
👉 Example: Store SQL password in Key Vault instead of pipeline JSON.
-

6. Explain Medallion Architecture in Databricks.

- **Bronze** = raw data.
 - **Silver** = cleaned & structured.
 - **Gold** = aggregated/BI-ready.
👉 Standard for scalable Lakehouse pipelines.
-

7. How do you optimize PySpark jobs?

- Broadcast small tables.
 - Cache reused datasets.
 - Repartition wisely.
 - Enable AQE.
👉 Example: Improved job from 2 hrs → 20 mins by partitioning on order_date.
-

8. What is the difference between Managed and External tables in Databricks?

- **Managed:** Databricks controls data + metadata (drop = delete).
 - **External:** Only metadata managed, data in ADLS.
👉 External preferred for enterprise data lakes.
-

9. Explain Delta Lake advantages.

- **ACID transactions**, schema enforcement, time travel, merges, compaction.
👉 Example: Handle late-arriving IoT data with Delta MERGE.
-

10. How do you handle schema evolution in Delta?

- Enable mergeSchema option.
 - Add/remove columns dynamically.
👉 Example: Added email column in customer table without pipeline failure.
-

11. What is Z-ordering?

- Multi-dimensional clustering in Delta.

- Optimizes queries filtering on multiple columns.
👉 Example: Queries on region + customer_id became 10x faster.
-

12. How do you monitor ADF pipelines?

- Use Monitor Hub.
 - Enable diagnostic logging to Log Analytics.
 - Set up alerts.
👉 Example: Teams notification on pipeline failure.
-

13. Explain Slowly Changing Dimensions (SCD).

- **Type 1:** Overwrite.
 - **Type 2:** New record with history.
 - **Type 3:** Add column for limited history.
👉 Example: Track customer address changes with Type 2.
-

14. How do you optimize file sizes in Data Lake?

- Compact small files.
 - Use columnar formats (Parquet/Delta).
👉 Example: Reduced query time by compacting 100k small files into 500 optimized files.
-

15. What is Data Vault modeling?

- **Hub** = business key.
 - **Link** = relationships.
 - **Satellite** = descriptive attributes.
👉 Example: Customer Hub + Order Hub + Customer-Order Link.
-

16. How do you handle late-arriving data?

- Use watermark in streaming.

- Merge into Delta with upserts.
👉 Example: Orders arriving 1 day late merged correctly.
-

17. Explain Adaptive Query Execution (AQE).

- Runtime optimizer in Spark.
 - Fixes skew, optimizes joins, reduces shuffle partitions.
👉 Example: Large shuffle join automatically replaced with broadcast.
-

18. How do you design a high-availability pipeline?

- Use retries, failover regions.
 - Stateless components with checkpoints.
👉 Example: Multi-region ADF pipelines for DR.
-

19. What is Unity Catalog?

- Governance layer in Databricks.
 - Centralized access, lineage, and auditing.
👉 Example: Restrict access to sensitive HR Delta tables.
-

20. How do you validate ETL pipeline output?

- Row count checks.
 - Hash totals.
 - Business rules validation.
👉 Example: Validate total sales = source system.
-

21. How do you troubleshoot Spark job failures?

- Check Spark UI for skew/shuffles.
 - Optimize joins & repartitions.
 - Review cluster memory configs.
-

22. Explain PolyBase.

- Allows Synapse to query external files in ADLS/Blob.
 - Supports high-speed parallel loads.
 - 👉 Example: Direct query parquet files for staging.
-

23. How do you reduce Databricks costs?

- Enable auto-termination.
 - Use spot VMs.
 - Compact small files.
 - 👉 Example: Saved 30% cost by using auto-termination on dev clusters.
-

24. What is Event Hub, and how does it integrate with Databricks?

- Event Hub = streaming ingestion service.
 - Databricks reads streams via Spark connector.
 - 👉 Example: Real-time clickstream analysis.
-

25. Explain Window functions in PySpark.

- Perform calculations across groups.
 - Examples: row_number(), rank(), lead(), lag().
 - 👉 Example: Get latest transaction per customer.
-

26. How do you design CI/CD for ADF & Databricks?

- Store pipelines in Git.
 - Deploy via DevOps pipelines.
 - Use ARM templates/APIs.
 - 👉 Example: Automated deployment from Dev → Prod.
-

27. What is the difference between OLTP and OLAP?

- **OLTP:** Real-time transactions.

- **OLAP:** Analytical, BI reporting.
👉 OLTP = banking, OLAP = customer analytics.
-

28. How do you implement error handling in ADF?

- Use On Failure activities.
 - Capture errors in log tables.
 - Trigger notifications.
👉 Example: Invalid file format logged into error table.
-

29. How do you manage schema drift in ADF?

- Enable schema drift in Mapping Data Flows.
 - Store unknown columns separately.
👉 Example: Handle new fields in JSON API data.
-

30. How do you design cost-optimized pipelines?

- Use serverless compute.
- Archive cold data.
- Optimize partitioning.
👉 Example: Use Synapse serverless for ad-hoc queries instead of dedicated pool.